

Marine reserves reduce risk of climate-driven phase shift by reinstating size- and habitat-specific trophic interactions

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Abstract. Spatial closures in the marine environment are widely accepted as effective conservation and fisheries management tools. Given increasing human-derived stressors acting on marine ecosystems, the need for such effective action is urgently clear. Here we explore mechanisms underlying the utility of marine reserves to reinstate trophic dynamics and to increase resilience of kelp beds against climate-driven phase shift to sea urchin barrens on the rapidly warming Tasmanian east coast. Tethering and tagging experiments were used to examine size- and shelter-specific survival of the range-extending sea urchin *Centrostephanus rodgersii* (Diadematidae) translocated to reefs inside and outside no-take Tasmanian marine reserves. Results show that survival rates of *C. rodgersii* exposed on flat reef substratum by tethering were approximately seven times (small urchins 10.1 times; large urchins 6.1 times) lower on protected reef within marine reserve boundaries (high abundance of large predatory-capable lobsters) compared to fished reef (large predatory lobsters absent). When able to seek crevice shelter, tag-resighting models estimated that mortality rates of *C. rodgersii* were lower overall but remained 3.3 times (small urchins 2.1 times; large urchins 6.4 times) higher in the presence of large lobsters inside marine reserves, with higher survival of small urchins owing to greater access to crevices relative to large urchins. Indeed, shelter was 6.3 times and 3.1 times more important to survival of small and large urchins, respectively, on reserved relative to fished reef. Experimental results corroborate with surveys throughout the range extension region, showing greater occurrence of overgrazing on high-relief rocky habitats where shelter for *C. rodgersii* is readily available. This shows that ecosystem impacts mediated by range extension of such habitat-modifying organisms will be heterogeneous in space, and that marine systems with a more natural complement of large and thus functional predators, as achievable within no-take reserves, will minimize local risk of phase shifts by reinstating size and habitat-specific predator–prey dynamics eroded by fishing. Importantly, our findings also highlight the crucial need to account for the influence of size dynamics and habitat complexity on rates of key predator–prey interactions when managing expectations of ecosystem-level responses within marine reserve boundaries.

Key words: *Centrostephanus rodgersii*; climate change; fishing; kelp beds; marine protected areas; predation; range extension; resilience; sea urchin; spatial refuge; Tasmanian temperate reefs; urchin barrens.

INTRODUCTION

Marine reserves are considered effective conservation and ecosystem-based management tools across the globe. Given the heavily impacted status of vast areas of the world's oceans (e.g., Myers and Worm 2003, Pauly et al. 2005, Berkes et al. 2006, Halpern et al. 2008), the need for effective ecosystem management intervention is urgently clear. Because marine reserves protect all component species from extractive activities, they arguably provide the most powerful local-scale tool for ecosystem-level protection, and, arranged as networks, offer a broad-scale approach to protect important marine ecosystem services against the increasing demands imposed by humans (reviewed by Gaines et al. [2010]).

While considerable variability is evident between locations, long-term changes to reef ecosystems inside marine reserves relative to adjacent fished sites clearly show that fishing has a major impact on the abundance, size, and fecundity of harvested species (reviewed by Halpern and Warner [2002], Halpern [2003], Edgar et al. [2009], and Lester et al. [2009]). In combination with unequivocal evidence arising from globally synthesized fishery-dependent data sets (e.g., Pauly et al. 1998, Jackson et al. 2001, Myers and Worm 2003, Berkes et al. 2006, Worm et al. 2006, Thurstan et al. 2010), long-term controlled experiments inside and outside marine reserves demonstrate that fishing, via altering trophic organization, can cause major change to the structure and functioning of marine ecosystems (reviewed by Edgar et al. [2009], Lester et al. [2009], and Babcock et al. [2010]). Ultimately, the establishment of marine reserves can demonstrate benefits beyond target species via the restoration of predator biomass resulting in cascading effects on lower trophic

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levels, including recovery of biogenic habitat-formers and concomitant increases in species richness (reviewed by Lester et al. [2009]). However, what remains largely unexplored is how or if indeed marine reserves can play a role in safeguarding local ecosystems against impacts of global change.

In addition to fishing, climate change is a key stressor impacting marine ecosystems globally and threatening major alterations to trophic organization via shifting species distributions (reviewed by Harley et al. [2006]). Given the range of current climate change projections (reviewed by Meehl et al. [2007]), impacts of species range shifts on trophic organization of marine ecosystems are also likely to be profound, particularly if and when species that influence underlying habitat distributions undergo shifts. While fishing and global climate change may independently pose major threats to ecosystem dynamics, interactions between these major stressors may lead to accelerated changes and unprecedented risk of phase shifts in coastal ecosystems (e.g., Dayton et al. 1998, Jackson et al. 2001, Hughes et al. 2003, Bellwood et al. 2004, Scheffer et al. 2005, Harley et al. 2006).

In eastern Tasmania, temperate coastal waters are warming at approximately four times the global ocean-warming average, representing one of the fastest rates of warming in the Southern Hemisphere (Ridgway 2007). This has driven poleward range extension of many marine species (Last et al. 2010, Johnson et al. 2011), including the ecologically important long-spined sea urchin *Centrostephanus rodgersii* (Ling et al. 2008, 2009b), which has commenced overgrazing of productive Tasmanian kelp beds (see Plate 1), leading to loss of local biodiversity (Ling 2008) and threatening key Tasmanian reef-based fisheries (abalone and lobster worth ~\$150 million Australian dollars per year before processing) (Johnson et al. 2011). Manifest as nocturnally grazed halos radiating from crevices used for daytime shelter (e.g., Andrew 1993; E. Flukes, unpublished data), *C. rodgersii* maintains barrens over ~50% of all near-shore rocky reefs within its native New South Wales range (Fig. 1), a pattern emerging at some sites in northeastern Tasmania in recent decades (Johnson et al. 2005; reviewed in Johnson et al. [2011]). Representing a catastrophic phase shift, there is a critical density of *C. rodgersii* required to initiate widespread overgrazing, with return to the kelp bed state rarely observed once this threshold density is passed (Andrew 1991, Andrew and Underwood 1993, Hill et al. 2003; reviewed by Ling et al. [2009a]). Given that recent winter warming has led to favorable temperatures for *C. rodgersii* larval development in eastern Tasmania (Ling et al. 2008, 2009b), there is considerable risk of continued urchin population increases and widespread overgrazing, as all rocky reefs on this coast are predicted to exceed the sea urchins 12°C larval development threshold by the year 2025 (Pecl et al. 2009).

Superimposed on the climate-driven arrival of *Centrostephanus rodgersii* in eastern Tasmania is heavy exploita-

tion of rocky reef ecosystems (e.g., Frusher 1997, Ling et al. 2009a). Long-term changes to reef ecosystems inside marine reserves relative to adjacent fished sites in eastern Tasmania show that fishing has a major impact on the abundance and size structure of harvested species on this coast (Edgar and Barrett 1999, Barrett et al. 2007, 2009, Edgar et al. 2009; reviewed in Babcock et al. [2010]). Showing strong local recovery within reserve boundaries, the lucrative spiny lobster *Jasus edwardsii* is a well-documented predator of sea urchins (Kelly et al. 2000, Shears and Babcock 2002, 2003, Pederson and Johnson 2006; reviewed by Babcock et al. [2010]) and known predator of *C. rodgersii* within the sea urchins' new Tasmanian range (Ling et al. 2009a). Demonstrating a strong effect of fishing on this species, projections from stock assessment models have estimated that the biomass of legal-sized spiny lobsters (CL $\geq$ 105 mm, female; 110 mm, male) on the east coast of Tasmania has recently been as low as 2–8% of unfished biomass (Frusher 1997). Importantly, only large lobsters ($\geq$ 140 mm carapace length; approximately $\geq$ 1.5 kg in mass), which demonstrate strong recovery within reserves, are physically capable of preying on emergent grazing size classes of *C. rodgersii* in eastern Tasmania (Ling et al. 2009a).

Understanding patterns of predation risk for *Centrostephanus rodgersii* within the range extension region is therefore essential to the identification of local reef conditions under which urchin population size will likely grow to exceed critical levels of grazing required to effect catastrophic phase shift. Focusing on the range extension region of southeastern Tasmania where *C. rodgersii* currently occurs in low abundance, we examine size and shelter-specific survival (and their interactive effects) of *C. rodgersii* following experimental translocation of the sea urchin to reefs inside marine reserves (high abundance of large predators) relative to adjacent fished reefs (low abundance of large predators). In combination with broad-scale correlative data on the distribution of the urchin across different reef habitats, experimental results are used to inform expectations of further overgrazing by the sea urchin given continued climate-driven population increases and heavy fishing of sea urchin predators within the range extension region.

MATERIALS AND METHODS

Study sites

Experiments were performed in two eastern Tasmanian regions where predator biomass has shown strong recovery inside marine reserves relative to adjacent reefs subject to exploitation (Edgar and Barrett 1999, Barrett et al. 2009). The Maria Island Marine Reserve (MIMR, Mercury Passage) was established in 1992 and had experienced 12 years of protection from fishing at the time of experimentation in 2004; the Crayfish Point Marine Reserve (CPMR, Derwent Estuary) was established in 1971 and had thus experienced 33 years of protection at time of experimentation (see Fig. 1). Experimental rocky reefs experienced moderate wave

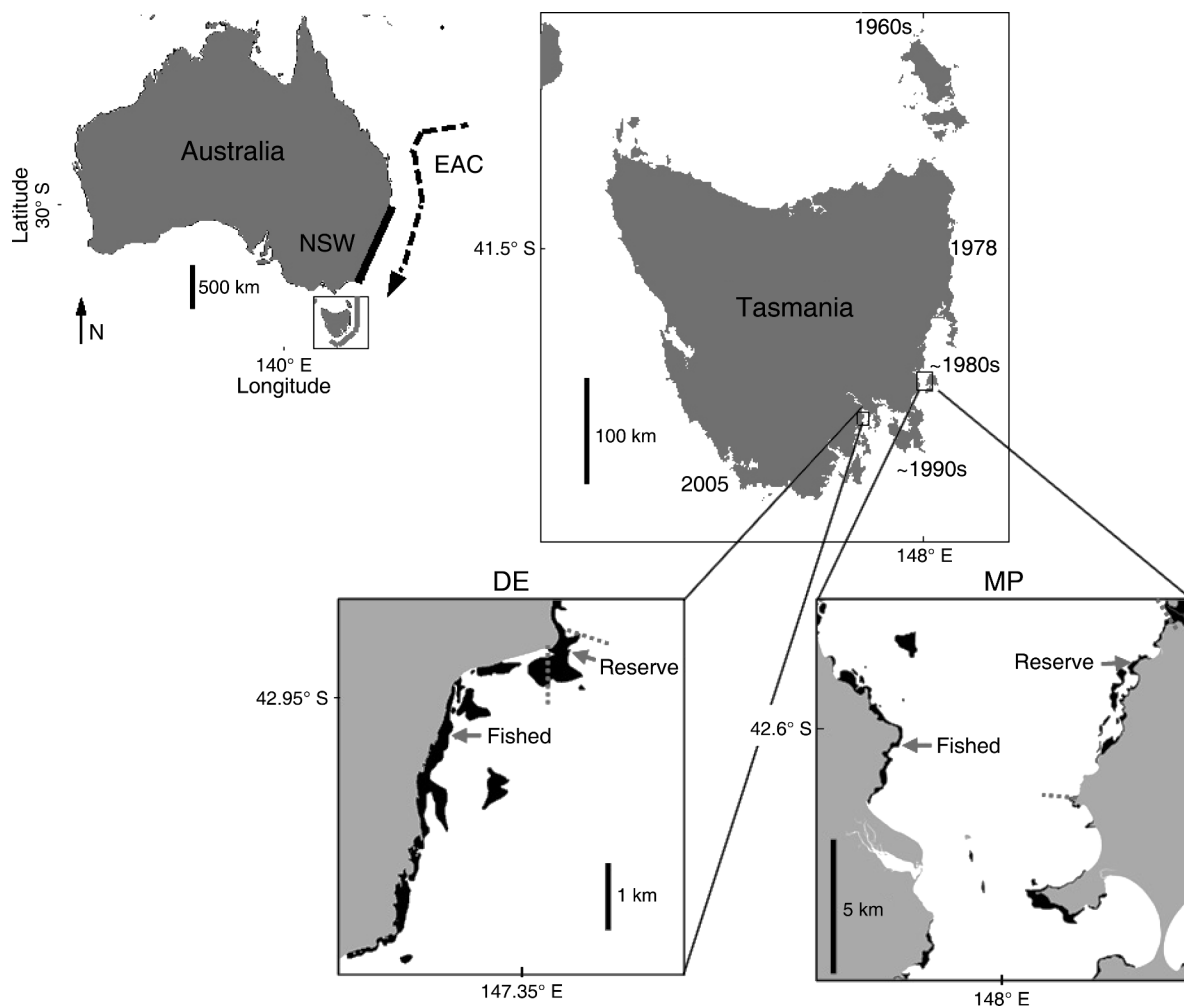


FIG. 1. Map of Tasmania showing locations and sites where predation experiments were undertaken. Accompanying the map of Australia is the directional influence of the East Australian Current (EAC); the former distribution of the sea urchin *Centrostephanus rodgersii* on the Australian mainland (solid line, after Andrew and Byrne [2007]); and its recent range expansion (thick gray line, after Johnson et al. [2005]). The expanded Tasmanian map shows the timing of first sightings of *C. rodgersii* along this coastline (after Johnson et al. [2005]; J. Valentine, *personal observation*). Expanded Derwent Estuary (DE) and Mercury Passage (MP) regions display experimental reefs (black polygons) inside marine reserves (CPMR and MIMR, respectively; boundaries are indicated by dotted gray lines) and nearby fished reefs (black areas) (Alum Cliffs and Johnson's Pt., respectively).

exposure, supported seaweed communities, and were all of high structural relief, being composed of ~50% cover of large (>1 m and <2.5 m diameter) and 50% cover of small (>0.2 m and <1 m diameter) boulders. Given low abundances of *C. rodgersii* within the immediate vicinity of the experimental reefs, sea urchins were collected from sites in northeastern Tasmania (near St. Helens), where *C. rodgersii* occurs in high abundance (~2.0 individuals/m²) and where all size ranges of the urchin were readily available to be transferred to experimental reefs in southeastern Tasmania, effectively simulating invasion by the sea urchin.

Abundance of predators inside and outside marine reserves

Field observations using remote video monitoring (aided by infrared lighting) revealed that large nocturnally

foraging spiny lobsters *Jasus edwardsii* (≥140 mm carapace length [CL]) and large diurnally foraging wrasse *Notolabrus tetricus* (≥350 mm total fish length [TFL]) are capable of preying on emergent *Centrostephanus rodgersii* (≥50 mm test diameter [TD]) that affect grazing in eastern Tasmania (Ling et al. 2009a). Large lobsters were identified as the chief predator capable of consuming all size classes of *C. rodgersii*, while relatively few wrasse predation events were witnessed and involved small urchins (≤60 mm TD), but only when the urchin was exposed from crevices during the daylight hours. For each experimental reef inside and outside the marine reserves, the abundance of predatory-capable spiny lobsters, *J. edwardsii* (≥140 mm CL) and the wrasse *N. tetricus* (≥350 mm TFL) were assessed from six replicate belt transects (50 × 4 m for lobsters and 50 × 10 m for wrasse, surveyed with a standardized search time of 30

minutes and 15 minutes for lobsters and wrasse, respectively). Predator size was estimated in situ using vernier calipers for lobster carapace length (to nearest 1 mm), while total fish length of wrasse was estimated using calibrated size-graded visual counts to the nearest 50 mm.

Survival of sheltering sea urchins

To examine the effect of sea urchin size on survival of *Centrostephanus rodgersii* sheltering inside and outside no-take marine reserves, individually tagged sea urchins were translocated onto reserve and fished reefs and resurveyed through time. At each of two reef sites inside and outside marine reserves, two experimental “census zones” 60 × 10 m were established into which 24 *C. rodgersii* in each of two size classes (small, 40–70 mm TD; large 80–120 mm TD) were translocated by placement of individuals within crevices (total of 48 sea urchins per census zone, 96 urchins per site, 384 tagged urchins in total) and monitored through time. Searches for tagged sea urchins were performed across each census zone (standardized as a 60-minute systematic diver search per zone) on eight resampling occasions (at 17, 37, 60, 86, 106, 131, and 182 days post-release) with the tag identity of urchins resighted alive recorded.

Tagging involved drilling two small holes through the test with a hypodermic needle (100 mm long by 1.25 mm diameter), threading a 150-mm length of monofilament line (0.45 mm diameter) through the needle, and threading a size 1 swivel-clip (8-mm clip opening) and numbered spaghetti tag over the monofilament before the line ends were crimped together with a leader sleeve (size 3). This method results in low mortality (<5%), and no signs of disease or necrosis were observed around the entry and exit points in the test. Because any mortality usually occurs within two days of the operation, tagged animals were monitored for several days and only healthy individuals were used in experiments. Furthermore, tagged individuals displayed normal behavior, and tracked individuals tagged during pilot trials retained readable tags in the field for over two years.

In addition, to test for possible differences in survivorship between tagged vs. untagged *C. rodgersii* in the presence of large predatory-capable lobsters, a 3 × 3 m cage with roof was deployed in the field housing a single male lobster (163 mm CL) with a total of 42 urchins (ranging from 50 to 115 mm TD) assigned equally to tagged and untagged treatments. This test was run at CPMR over two months, from April to May 2005. We considered the tagging procedure to have very small, if any, effect on the likelihood of lobster predation, which was corroborated by observation that equal numbers of tagged ($n = 16$) and nontagged urchins ($n = 16$) were consumed when caged with a large predatory lobster in the field over two months.

Predation risk of exposed vs. sheltering sea urchins

The effect of urchin size and crevice shelters on survival of *Centrostephanus rodgersii* was examined

inside and outside the MIMR by comparing the survival of tagged (able to seek crevice shelter) with survival of tethered sea urchins (unable to seek crevice shelter). Tethering of *C. rodgersii* involved the same protocol as for tagging (see *Survival of sheltering sea urchins*), but with the inclusion of a swivel-clip (8 mm clip opening) threaded over the monofilament before the line ends were crimped together with a leader sleeve. As for tagging, this method resulted in low mortality (<5%), with no signs of disease or necrosis observed around the entry and exit points in the test. Individuals tethered in this manner were capable of surviving in the field for >100 days. Further note that *C. rodgersii* with small drill holes in the test, as per the tethering and tagging field experiments, did not elicit elevated rates of attack from lobsters under aquarium conditions.

Based on limited availability of flat rock surfaces in each of two experimental zones (30 × 10 m and separated by a distance of 50 m) inside and outside the MIMR (Mercury Passage), tethered sea urchins were arranged in 16 tethering arrays. Each tethering array consisted of a flat rock surface (i.e., boulder top) devoid of crevice refuges for urchins, which was cleared of macroalgae to avoid any potential impact of swell-driven macroalgal whiplash on urchins (e.g., Ling and Johnson 2009) and was scrubbed down to bare rock, allowing tenacious tube feet attachment as occurs naturally within crevices where shading naturally restricts macroalgal growth. For each of the 16 tethering arrays in each experimental zone, three individuals of two size classes (small, 40–70 mm TD; large, 80–120 mm TD) were clipped along a 2-m length of 6 mm diameter galvanized steel chain anchored firmly with steel pegs on either side of the rock platform, amounting to 24 urchins of each size class tethered per experimental zone (a total of 182 tethered *C. rodgersii* across two experimental zones inside and outside MIMR). Sea urchins were clipped to the chain at equal spacing, ensuring that individuals could not cluster together.

Survival of tethered sea urchins was assessed inside and outside the MIMR on nine occasions (approximately every second week) over a 100-day period by examining each tether array and recording the status of the attached sea urchin as dead or alive, with no replacement of dead individuals. Tethers with missing urchins or the remains of sea urchins attached, or remains nearby, were recorded as mortality events. Over the duration of the experiment, escapees were recorded, as some tether clips corroded due to prolonged exposure (17 in marine reserve; 21 on fished reef). Escapees occurred approximately evenly across all experimental zones and were factored out of the analysis. Escapees were easily identifiable, as they still had their individually coded tags intact, even though swivel clips had corroded from the tethering chain attachment point. In all cases, escapees were found to be seeking shelter in crevices immediately adjacent (within 2 m) of the tethering array where originally attached.

*Habitat-specific patterns of grazing
within the extended range*

The distribution of *Centrostephanus rodgersii* barrens across different rocky reef habitat types was assessed for eastern Tasmania by examining cover of the grazed area in 5×1 m quadrats that were laid haphazardly across different reef substrata. Assessed in situ by divers, quadrats were surveyed contiguously along each of 12 transect lines set perpendicularly from the shore (~6 m depth to a maximum of 18 m depth) at nine eastern Tasmania sites open to fishing and separated by ~20 km, ranging in latitude from 40°55' S to 43°10' S (see Johnson et al. 2005). Each quadrat was assigned to a particular habitat category based on the dominant substratum type (i.e., that substratum type contributing >60% of the quadrat by area). Quadrats for which there was no dominant substratum type, or for which the dominant substratum was sand, were excluded from the analysis. Substratum was classified as either flat rock shelves (>5 m effective diameter); very large boulders (>2.5 m and <5 m diameter); large boulders (>1 m and <2.5 m diameter); small boulders (>0.2 m and <1 m diameter); or other hard substrata (<0.2 m diameter).

DATA ANALYSIS

The abundance of large predatory-capable lobsters and wrasse were compared inside and outside the marine reserves using mixed effects two-way analysis of the factors “Reserve” status (protected vs. fished; fixed effect) and “Site” (Mercury Passage vs. Derwent Estuary; random effect). ANOVAS were undertaken using SAS version 9.1 (SAS 2004), with data for ANOVA checked for conformity to assumptions of homoscedasticity and normality. Where data were heteroscedastic, the transformation to stabilize variances was determined by the relationship between group standard deviations and means (Draper and Smith 1981). The appropriate transformation for each variable is expressed in the ANOVA results in terms of the untransformed variable Y .

Multiple resurvey of tagged urchins through time yielded individual “encounter histories” (individuals scored as either alive and present “1” or absent “0” at each resurvey period) enabling mark–recapture analysis according to the Cormack–Jolly–Seber (CJS) “recaptures only” routine available in the Program MARK software (White and Burnham 1999). For CJS, the number of individuals resighted alive on subsequent sampling occasions is a function of two probabilities: the probability of survival (denoted ϕ), and the probability that conditional on surviving, the surviving individual is encountered (p). Program MARK uses Maximum Likelihood estimation to derive estimates of the parameters ϕ and p that maximize the likelihood of witnessing the observed frequency of individuals across different encounter history scenarios. Our experimental design allowed modeling of the contributions of “group” and “time” (i.e., sampling occasions, $n = 8$) to apparent

daily survival and encounter probabilities. “Group” included all combinations of two urchin size classes at four reef sites, two sites inside and two sites outside marine reserves, giving $n = 8$ unique groups. (Census zones were pooled within each site to minimize the number of estimable parameters for this analysis.)

The starting point of the analysis is to confirm that the saturated CJS model (all parameters included) adequately describes variability in the data, i.e., that survival probability (ϕ) and encounter probability (p) depend on both group (g) and time (t), expressed formally as $\phi(g \times t)p(g \times t)$, with saturated model fit examined using 1000 bootstraps within the goodness-of-fit (GOF) routine available in Program MARK. The most parsimonious CJS model (based on Akaike’s Information Criterion) is then identifiable by excluding parameters from the saturated model that cannot be justified by the data. In this approach there is no null hypothesis or formal test of significance, but rather the concept is one of “best inference” given the data and a set of a priori models (Lebreton et al. 1992).

To examine patterns in survival and encounter estimates generated by the best-supported mark–recapture model, the factors of “reserve” status (protected vs. fished) and urchin “size” (small vs. large) were tested using fixed-effects two-way ANOVA. Estimates of mortality (i.e., $1 - \text{survival}$) and encounter probability of small and large *Centrostephanus rodgersii* were then compared against the density of large predatory-capable lobsters and wrasse at each site using simple linear regression.

Survival estimates of exposed (tethered) vs. sheltering (tagged) sea urchins were compared using the two experimental zones inside and outside MIMR as replicates in a fixed-effects three-way ANOVA testing the factors of “Reserve” status (protected vs. fished), urchin size (small vs. large), and shelter (exposed vs. shelter) on sea urchin survival at ~90 days post-release of the sea urchins (i.e., the temporal extent in overlap of tether and tag trials). To estimate tagged urchin survival at the ~90 day time step, and for graphical comparison with tethered urchins over all time steps, the time-dependent saturated CJS model, $\phi(g \times t)p(g \times t)$, was used to generate survival of tagged urchins in each size class, in each experimental zone, at each site inside and outside the MIMR.

Patterns in percentage cover of barrens habitats across different reef substratum types was analyzed using an unbalanced one-way ANOVA. Where required, multiple range tests were conducting using the Ryan–Einot–Gabriel–Welsch (REGWQ) procedure.

RESULTS

Abundance of predators inside and outside marine reserves

The abundance of large spiny lobsters (*Jasus edwardsii* ≥140 mm CL) capable of preying on emergent *Centrostephanus rodgersii* under natural field conditions was significantly higher inside marine reserves than at fished sites where such individuals were functionally absent (Fig. 2a; two-way ANOVA, transformation is

$\log(Y)$, effect of “Reserve” (fixed), $F_{1,1} = 453.83$, $P = 0.029$; “Site” (random), $F_{1,20} = 0.11$, $P = 0.75$; “Reserve $\times$ Site” (random), $F_{1,20} = 0.11$, $P = 0.75$). In contrast, the pattern in abundance of large wrasse (*Notolabrus tetricus* ≥ 350 mm TFL), also previously observed to be capable of preying on *C. rodgersii* (albeit small urchins when exposed during daylight), was not significantly different between experimental reefs inside and outside marine reserves (Fig. 2b; two-way ANOVA, transformation is $\sqrt{Y}$, “Reserve” (fixed), $F_{1,1} = 0.37$, $P = 0.65$; “Site” (random), $F_{1,20} = 0.67$, $P = 0.42$; “Reserve $\times$ Site” (random), $F_{1,20} = 0.37$, $P = 0.38$).

Other demersal carnivorous and/or omnivorous fishes also showed variable patterns of abundance inside and outside marine reserves, and the total density of demersal carnivorous/omnivorous fishes (including *N. tetricus*) inside marine reserves was 3.85 ± 0.92 individuals/100 m² (mean $\pm$ SE), compared to 4.77 ± 1.17 individuals/100 m² at the fished sites.

Resighting and survival of sheltering sea urchins

Multiple resurvey of individually tagged *Centrostephanus rodgersii* revealed a consistent decline in resighting rates of the sea urchin inside marine reserves relative to a high and stable rate of resightings of tagged individuals at sites open to fishing (Fig. 3a). Resighting rates at the fished sites stabilized at $\sim 70\%$ of the total tagged urchins released, whereas resighting rates inside the marine reserves declined sharply before stabilizing at $\sim 25\%$ of the tagged urchins released to the reef (Fig. 3a). Patterns of resighting were broadly similar for small and large sea urchins inside marine reserves (cf. dashed lines in Fig. 3b, c). However, outside marine reserves resighting rates for large urchins were generally high compared to small sea urchins over the study period (cf. solid lines in Fig. 3a, b). Generated from patterns of live resightings of tagged individuals, modeling of encounter histories with the saturated time-dependent and size-specific CJS model, $\phi(g \times t)p(g \times t)$, indicated satisfactory goodness-of-fit (as determined by nonsignificant “lack of fit”, $P = 0.12$). The model reduction process indicated that the most parsimonious model contained survival (ϕ) and encounter (p) probabilities as a function of experimental group (g) only, $\phi(g)p(g)$, i.e., groups based on all combinations of small and large urchins, inside and outside two marine reserves, giving eight groups in total.

Estimates of daily survival probability generated by the best supported “group only” model, $\phi(g)p(g)$, i.e., integrating across the full temporal extent of the data (182 days), revealed reduced survival of *Centrostephanus rodgersii* inside the marine reserves relative to reefs open to fishing (Fig. 4a, see ANOVA result, Table 1: Survival). The proportion of *C. rodgersii* populations experiencing mortality (i.e., $1 - \text{survival}$) by the conclusion of the tagging experiment was 3.3 times higher inside marine reserves (0.69) compared to fished reefs (0.21), with mortality of small urchins 2.1 times higher inside reserves (0.63) than fished reef (0.30), and mortality of large urchins 6.4 times higher inside reserves

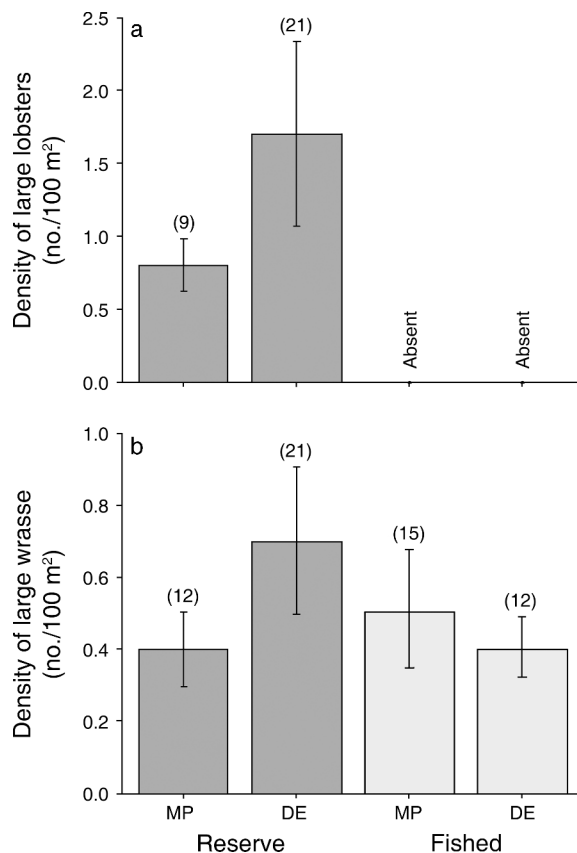


FIG. 2. Abundance (mean $\pm$ SE) of sea urchin predators observed to be capable of preying on *Centrostephanus rodgersii* at reserve and fished sites in Mercury Passage (MP) and Derwent Estuary (DE). (a) Local abundance of large spiny lobsters, *Jasus edwardsii* (>140 mm carapace length [Ling et al. 2009b]), as assessed by diver belt transects 4×50 m ($n = 6$) at the sea urchin tethering and tagging experimental area. (b) Abundance of large blue-throated wrasse, *Notolabrus tetricus* (>350 mm total fish length), as assessed by diver belt transects 10×50 m ($n = 6$) at each experimental reef (see Results). Numbers in parentheses indicate the total number of individuals observed on transects at each site.

(0.74) than fished reef (0.12). Estimates of encounter probabilities were also consistently lower inside the marine reserves relative to reefs open to fishing (Fig. 4b, see ANOVA result, Table 1: Encounter). While the effect of sea urchin size on survival of tagged sea urchins was not significant (Table 1: Survival), size did have a significant effect on encounter probability (Table 1: Encounter). Importantly, for both survival and encounter probabilities, significant interaction effects were detected between reef status and sea urchin size (Table 1). Inside marine reserves, small urchins displayed greater crypsis (i.e., lower encounter probability) and demonstrated higher survival compared to large urchins that were more exposed (i.e., higher encounter probability) and experienced lower survival (cf. Fig. 4a, b). In contrast, for the fished reef, large urchins not only displayed greater exposure but also demonstrated greater survival relative to small urchins (cf. Fig. 4a, b).

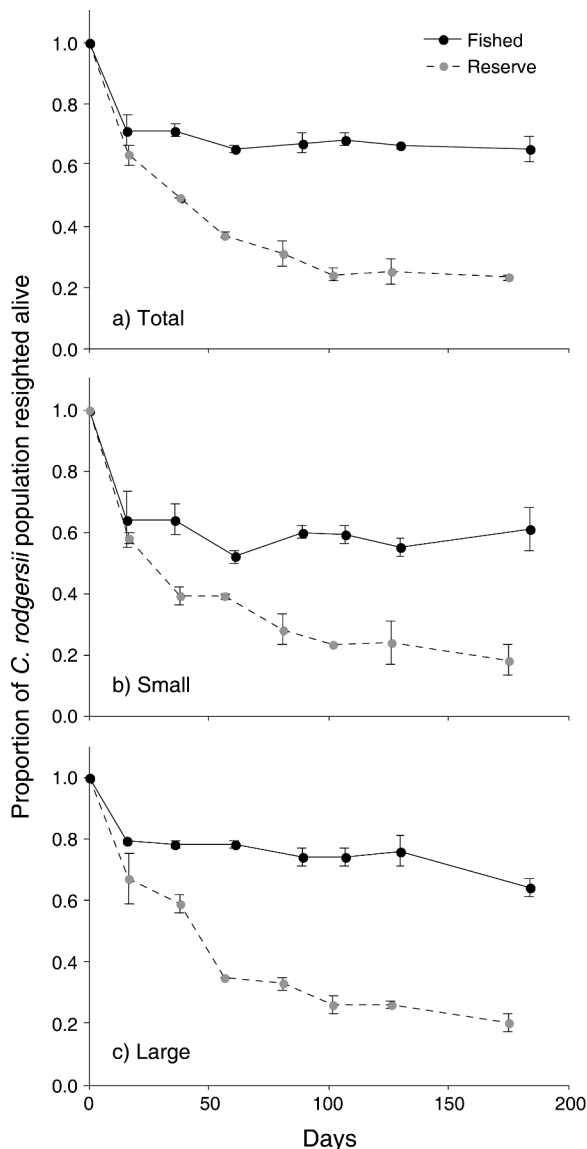


FIG. 3. Proportion (mean $\pm$ SE) of tagged *Centrostephanus rodgersii* resighted alive through time at experimental reefs inside (dashed lines) and outside (solid lines) marine reserves in Mercury Passage and the Derwent Estuary. Graphs show (a) total sea urchins (small and large combined); (b) small sea urchins (40–70 mm test diameter, TD); (c) large sea urchins (80–120 mm TD).

Examination of apparent daily mortality of *Centrostephanus rodgersii* with respect to the density of large lobsters revealed evidence for a threshold-type response between urchin mortality and the presence of predatory-capable lobsters for both small and large sea urchins (Fig. 5a). In addition, there was a marked decline in encounter probability of both small and large sea urchins in the presence of supralegal-sized lobsters (Fig. 5a). With respect to the density of predatory-capable wrasse, urchin survival and encounter probability appeared largely independent of the density of

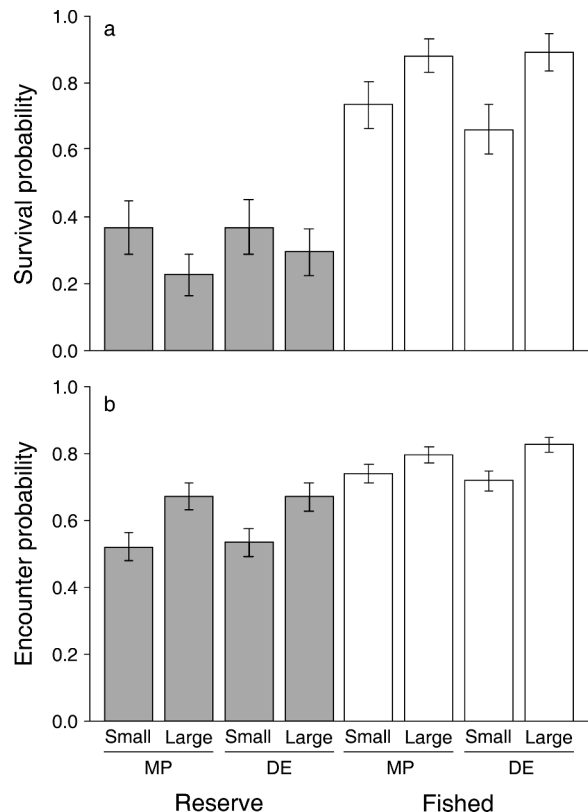


FIG. 4. (a) Size-specific survival (displayed as mean daily survival rate projected one-half year) and (b) encounter probabilities (mean proportion of the population sighted) for *Centrostephanus rodgersii* on experimental reefs inside (Reserve) and outside (Fished) marine reserves, as estimated (mean $\pm$ SE) using the best-supported Cormack-Jolly-Seber model; apparent daily survival and encounter probabilities are dependent on the experimental group only. Size classes of *C. rodgersii* (small, 40–70 mm TD; large, 80–120 mm TD) and locations (MP, Mercury Passage; DE, Derwent Estuary) are shown for all experimental sites.

TABLE 1. ANOVA results for survival and encounter probabilities for tagged small and large *Centrostephanus rodgersii* sea urchins (body size factor, fixed) inside and outside protected areas (reserve factor, fixed) in Mercury Passage and the Derwent Estuary, Tasmania, Australia.

Source	df	MS	F	P
Survival				
Reserve	1	0.0000530	176.67	0.0002
Body size	1	0.0000002	0.66	0.4607
Reserve $\times$ body size	1	0.0000544	18.14	0.0131
Error	4	0.0000003		
Encounter				
Reserve	1	0.0580048	262.46	<0.0001
Body size	1	0.0256677	116.14	0.0004
Reserve $\times$ body size	1	0.0020146	9.12	0.0392
Error	4	0.0002210		

Note: Significant *P* values are in boldface type.

large wrasse for both small and large sea urchins (Fig. 5b). Nonetheless, if only considering small sea urchins, greatest crypsis (i.e., lowest encounter probability) and highest mortality rate were both observed where density of large wrasse was highest (Fig. 5b). However, it is important to note that this data point corresponds to the place where lobster density was also greatest. Hence this pattern is unlikely to be an effect of wrasse alone.

Survival of exposed vs. sheltering sea urchins

The proportion of exposed (i.e., tethered) *Centrostephanus rodgersii* surviving through time declined rapidly inside marine reserves relative to a subtle decline observed on reef open to fishing (Fig. 6a). Showing a strong preference by predators for small urchins within reserve boundaries, ~50% of all small urchins were preyed upon during the first five days, whereas only 2% of large urchins were consumed over the same period when both size classes were exposed on flat rock surfaces (cf. Fig. 6b, c). Relative to exposed sea urchins, both small and large urchins demonstrated higher survival when able to seek crevice shelters (cf. Fig. 6a, b). When analyzed at the full temporal extent of the data set (i.e., at 90 days), there was a significant interaction term between “Reserve” status and “Shelter,” indicating that shelter inside marine reserves is more important to sea urchin survival than on fished reefs (Table 2). Indeed comparison of tagged and tethered urchin survival inside and outside reserve boundaries revealed that shelter was overall approximately four times more important to *C. rodgersii* survival (small urchins 6.3 times; large urchins 3.1 times) on reserve reef relative to fished reef (Table 3).

Substratum-specific patterns of grazing

Data spanning different rocky reef substrata across eastern Tasmania show that *Centrostephanus rodgersii* barrens predominantly form on boulder reef habitats (ranging from >0.2 to <5.0 m diameter) where crevice shelter is readily available (Fig. 7). Flat rock substratum, where sea urchins must remain exposed on the reef surface, was significantly less impacted by grazing across eastern Tasmania compared to boulder reef habitats (one-way ANOVA, $F_{3,1524} = 5.51$, $P < 0.001$, see REGWQ groupings in Fig. 7).

DISCUSSION

Shelter-specific dynamics inside and outside marine reserves

Expanding on previous experiments showing relatively high rates of predation on sea urchins within temperate marine reserves (e.g., Sala and Zabala 1996, Cole and Keuskamp 1998, Shears and Babcock 2002, Guidetti 2006, Pederson and Johnson 2006), our results clearly demonstrate the ability of no-take reserves to reinstate size- and habitat-specific trophic dynamics otherwise missing from reefs on the rapidly warming Tasmanian east coast. In the presence of large predato-

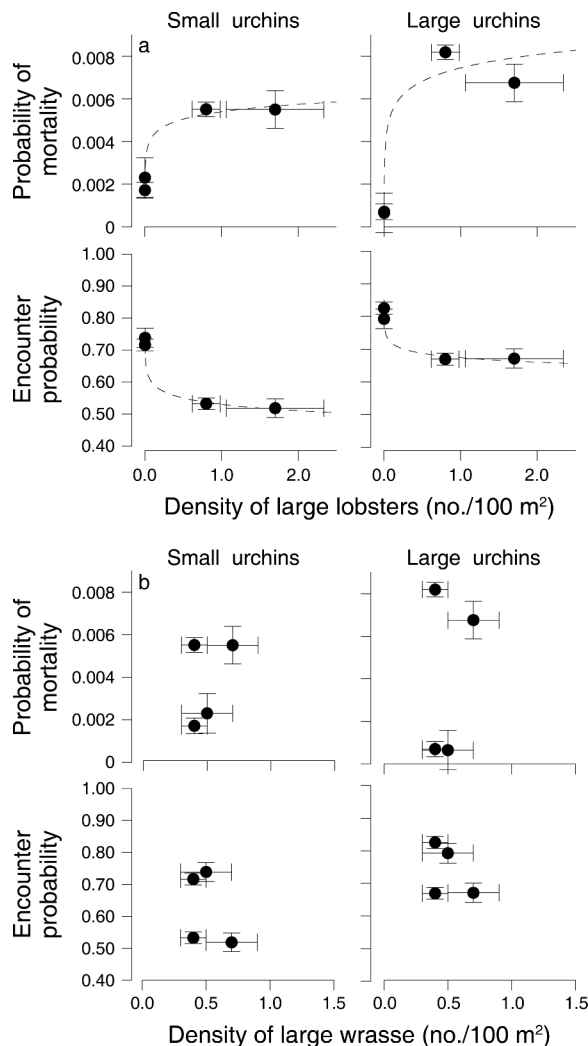


FIG. 5. Daily mortality ($1 - [\text{apparent daily survival}]$) and encounter probabilities (mean proportion of the population sighted) for *Centrostephanus rodgersii* as a function of (a) abundance of large lobsters (≥ 140 mm carapace length) and (b) abundance of large wrasse (≥ 350 mm total fish length), expressed for each predator as individuals per 100 m², for experimental sites inside and outside marine reserves. On all axes, data are shown as mean $\pm$ SE. For panel (a), mortality increased with increasing abundance of large lobsters (relationship best described by logarithmic fit [dashed lines], indicating a threshold type response: $y = 0.0005 \ln(x) + 0.008$, $R^2 = 0.98$; and $y = 0.0009 \ln(x) + 0.01$, $R^2 = 0.95$ for small and large urchins, respectively). Encounter probability decreased with increasing abundance of large lobsters: $y = -0.029 \ln(x) + 0.40$, $R^2 = 0.99$; and $y = -0.020 \ln(x) + 0.58$, $R^2 = 0.96$ for small and large urchins, respectively). For panel (b), *C. rodgersii* mortality and encounter probability appeared independent of large wrasse abundance.

ry-capable lobsters inside reserves, predation rates on the range-extending *Centrostephanus rodgersii*, when exposed on flat reef substratum, were approximately seven times higher than that observed on adjacent fished reef where predatory lobsters were functionally absent

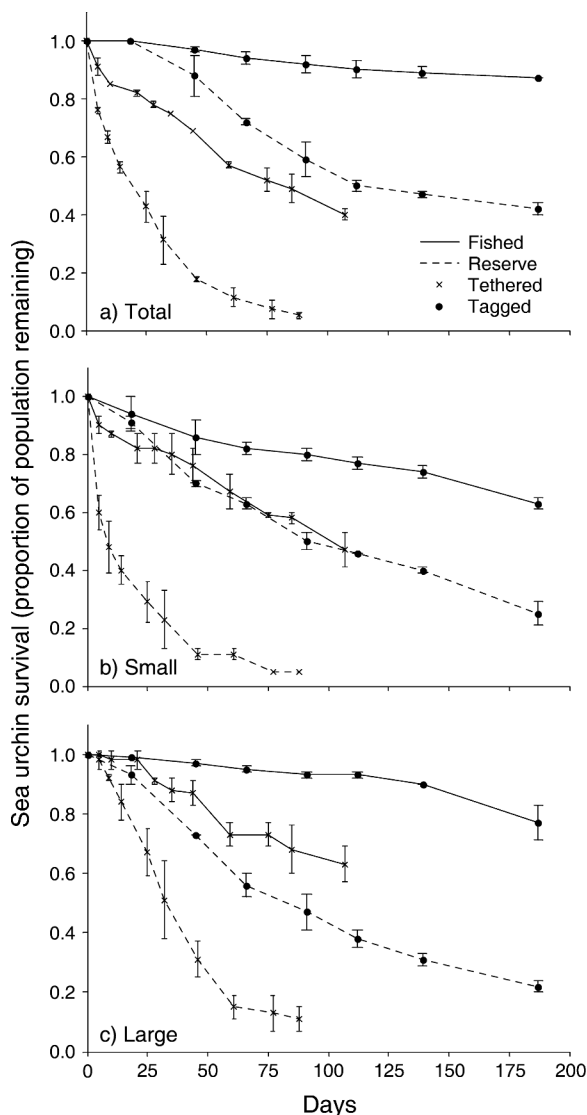


FIG. 6. Survival of *Centrostephanus rodgersii* with access to shelter (tagged urchins) and without shelter (tethered urchins) inside (dashed lines) and outside (solid lines) the Maria Island Marine Reserve. Graphs show (a) total sea urchins (small and large combined), (b) small sea urchins (40–70 mm test diameter, TD), and (c) large sea urchins (80–120 mm TD).

(Fig. 6). When able to utilize protective crevice shelters, populations of tagged *C. rodgersii* demonstrated greater overall survival compared to exposed urchins, but also showed increased mortality (3.3 times higher) when translocated to reefs inside reserves relative to fished reefs. By demonstrating a positive effect of crevice shelters on sea urchin survival, our findings extend prior inference regarding the importance of crevice shelters to *C. rodgersii* and its grazing mode (Andrew 1993). Furthermore, our work broadens the findings of smaller-scale field and laboratory experiments in the northern hemisphere, demonstrating the importance of shelter to the survival of juvenile sea urchins when in the

TABLE 2. ANOVA results examining reserve, body size, and shelter effects (all fixed effects) on the survival of *Centrostephanus rodgersii* after 90 days inside and outside the Maria Island Marine Reserve.

Source	df	MS	F	P
Reserve	1	0.7225	98.47	<0.0001
Body size	1	0.0342	4.66	0.0628
Shelter	1	0.5112	69.67	<0.0001
Reserve $\times$ body size	1	0.0012	0.17	0.6936
Reserve $\times$ shelter	1	0.0650	8.86	0.0177
Body size $\times$ shelter	1	0.0009	0.12	0.7352
Reserve $\times$ body size $\times$ shelter	1	0.0001	0.01	0.9099
Error	8	0.0073		

Notes: Significant *P* values are in boldface type.

presence of predators (e.g., Scheibling and Hamm 1991, Hereu et al. 2005, Scheibling and Robinson 2008).

Size-specific sheltering dynamics

Mediated by the availability of shelter for sea urchins of a given size, our experiments revealed a predator-driven size dynamic controlling *Centrostephanus rodgersii* survival. Showing a strong preference by predators for small urchins within reserve boundaries, ~50% of all small urchins were preyed upon during the first five days when exposed on flat rock surfaces, whereas only 2% of large exposed urchins were consumed over the same period (cf. Fig. 6b, c). Furthermore, a predatory switch to larger urchins was apparent once the abundance of small urchins had declined such that by final sampling (~90 days) small urchins had experienced an overall 10 times and large urchins a 6 times higher mortality inside reserve boundaries relative to fished reef (Table 3, Fig. 6b, c). In comparison to exposed patterns at 90 days, small urchins, when capable of seeking shelter, showed

TABLE 3. Comparison of survival rates at ~90 days for tagged (with shelter) and tethered (without shelter) *Centrostephanus rodgersii*, by size, inside and outside the Maria Island Marine Reserve.

Source	Reserve	Fished	Reserve factor
Small urchins			
Tagged	0.50	0.80	1.60
Tethered	0.05	0.51	10.10
Shelter factor	9.97	1.58	6.29
Large urchins			
Tagged	0.47	0.93	1.99
Tethered	0.11	0.64	6.10
Shelter factor	4.46	1.46	3.06
Total urchins			
Tagged	0.48	0.87	1.79
Tethered	0.08	0.57	7.39
Shelter factor	6.24	1.51	4.12

Notes: Shelter factor indicates the number of times that tagged urchin survival was greater than tethered urchin survival. Reserve factor indicates the number of times that urchin survival was greater on fished relative to reserve reef. Numbers in boldface type indicate the factor by which shelter is more important to urchin survival within reserve boundaries as compared with fished reef.

lower mortality, only 1.6 times higher mortality inside marine reserves, relative to large urchins, which showed a 2 times higher mortality when tagged and able to seek crevice shelters inside reserves (Table 3: Small, Fig. 6b). This seemingly counterintuitive result is explained by the small size class of urchins demonstrating lower resighting rates and concomitant higher survival within marine reserves relative to large urchins, which were more visible and therefore more exposed to a greater risk of predation (Fig. 4a, b). Conversely, on fished reefs where large-lobster abundance and thus predation risk was low, small urchins were also more cryptic than large urchins, but here large urchins displayed greater survival than small urchins despite being more exposed on the reef surface (Fig. 4). Ultimately, shelter was ~6.3 times more important to small urchins within reserve boundaries relative to fished reef; whereas shelter was of less importance for large urchins, but was still ~3 times more important to survival of large urchins when occurring within reserve boundaries relative to fished reef.

Complex size and shelter dynamics have also been described for the native short-spined sea urchin (*Heliocidaris erythrogramma*) in eastern Tasmania, whereby small, exposed individuals were also more vulnerable to predators than large urchins (Pederson and Johnson 2006). But as revealed for *Centrostephanus rodgersii* in the current study, the reverse was true when both size classes of *H. erythrogramma* were free to seek shelter within a crevice structure. Inferred to be a result of limiting crevice space as juvenile urchins outgrow microhabitat refuges, intermediate-sized urchins are reported to display greatest predation risk, which is supported by observations of bimodal size distributions on reefs where urchin populations are subjected to predation (e.g., Tegner and Dayton 1981, Tegner and Levin 1983, Sala and Zabala 1996, Shears and Babcock 2002, Hereu et al. 2005). Indeed, only large sea urchins with an apparent size refuge are reported to remain exposed on reef surfaces in the presence of predators (Tegner and Dayton 1981, Cowen 1983, Shears and Babcock 2002, Guidetti 2006). Such size specificity has clear implications for urchin population dynamics. In the presence of functional predators within reserves, sea urchin populations will experience predator-driven mortality over a greater portion of their life span relative to fished reefs, where average predator size is small and where urchins obtain a size refuge at a relatively younger age and/or smaller size (e.g., Cowen 1983, Tegner and Levin 1983, McClanahan and Muthiga 1989, Shears and Babcock 2002, 2003, Pederson and Johnson 2006).

Substratum-specific risk of barrens formation

Consistent with other studies showing that availability of crevice shelter is an important determinant of sea urchin distribution and abundance (e.g., Vance and Schmitt 1979, Tegner and Dayton 1981, Carpenter 1984,

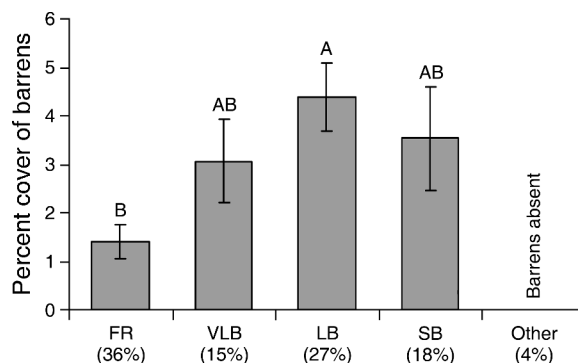


FIG. 7. Distribution of *Centrostephanus rodgersii* barrens (means ± SE) across different rocky reef substratum types in eastern Tasmania (6–18 m depth). Categories are flat rock shelves (FR, >5 m effective diameter); very large boulders (VLB, >2.5 m and <5 m diameter); large boulders (LB, >1 m and <2.5 m diameter); small boulders (SB, >0.2 m and <1 m diameter); and “Other,” which includes hard reef substrata <0.2 m diameter. Uppercase letters indicate that REGWQ (Ryan-Einot-Gabriel-Welsch range test) groupings were significantly different at $\alpha = 0.05$. Relative percentages of each reef substratum type sampled across eastern Tasmania are shown in parentheses below the axis; $N = 1529$ quadrats.

McClanahan and Kurtis 1991, Andrew 1993, Hereu et al. 2005), results from eastern Tasmania show that overgrazing by *Centrostephanus rodgersii* is indeed centered on boulder fields providing ample crevice protection (Fig. 7). As observed by remote infrared cameras, predation by lobsters occurs exclusively at night when *C. rodgersii* emerges from crevice shelters and is more prone to being inverted and attacked via the vulnerable peristomial opening (mouth) on the sea urchins’ less protected underside (Ling et al. 2009a). As observed across several taxa, sea urchins chiefly mitigate predation risk by resisting inversion and exposure of the vulnerable peristomial opening to predators (e.g., Tegner and Dayton 1981, Tegner and Levin 1983, Andrew and Underwood 1989, Andrew 1993, Guidetti and Mori 2005, Gianguzza et al. 2010). Hence an ability to brace within crevices, in combination with a spiny defense, appear the key determinants of sea urchin survival in the presence of predators.

While the predominant mode of barrens formation by *Centrostephanus rodgersii* hinges on nocturnal grazing excursions from their daytime shelters (e.g., Andrew 1993; E. Flukes, unpublished data), shelter is, however, not obligatory for barrens formation as barrens also occur, albeit less frequently, across featureless flat-rock substratum (see Fig. 7). Under such situations *C. rodgersii* is observed to remain exposed on the reef surface both night and day, but only when population density is high and the local risk of predation deemed low (Andrew and O’Neill 2000; reviewed by Andrew and Byrne [2007]). Such high-density populations and widespread barrens ($\geq 10\,000\text{ m}^2$) effecting wholesale change in reef ecology are currently pronounced along the northeast to mid-eastern Tasmanian coastline,

predominantly in depths >18 m (Johnson et al. 2005). But while the extent of barrens is patchily distributed and is currently low at the scale of the sea urchins' entire range extension region (<5% barrens cover of reefs; i.e., Fig. 7), our results suggest differential risk of barren expansion and associated ecological consequences across rocky reef habitat types. Indeed, based on the distribution of reef substratum types and projected coastal warming, *C. rodgersii* barrens habitat could potentially expand to account for ~50% of reefs in eastern Tasmania (Johnson et al. 2005), as is observed within the urchins' native New South Wales range (Andrew and O'Neill 2000). Given continued intensive fishing of sea urchin predators in eastern Tasmania, our results show that the survival of urchins and therefore risk of barrens formation will be greatest for boulder substratum types, but that the risk of barrens formation will be disproportionately high on open reef surfaces in the absence of functional predators.

Maximizing local predation risk to sea urchins

Consistent with a high frequency of lobster attacks on *Centrostephanus rodgersii*, as previously reported in Ling et al. (2009a) (i.e., lobsters accounted for 24 of 26 predation events captured by remote video monitoring), regressions of urchin mortality vs. density of large predatory-capable lobsters and wrasse in the current study indicate that large lobsters, not wrasse, drive patterns in *Centrostephanus rodgersii* survival inside and outside marine reserves (Fig. 5). Examining predation on the short-spined sea urchin (*Heliocidaris erythrogramma*), experiments by Pederson and Johnson (2006) also revealed a greater local predatory effect by lobsters (*Jasus edwardsii*) on urchins relative to the wrasse *Notolabrus tetricus* in eastern Tasmania. While we observed few predation events by labrids on *C. rodgersii*, all involved small urchins exposed from crevice shelters during daylight hours being consumed by large *Notolabrus tetricus* >350 mm in length (Ling et al. 2009a). Thus, notwithstanding the seemingly limited effect of predation by wrasse in eastern Tasmania, as for lobsters, sized-based shifts in dietary breadth are also known for labrids (e.g., Cowen 1983, Gillanders 1995, Hamilton et al. 2011). Notably in the case of small *C. rodgersii*, the lowest resighting rate plus a high mortality rate were both observed where density of large wrasse, but also lobsters, was highest (i.e., the CPMR site). Given high rates of mortality for exposed *C. rodgersii* within reserve boundaries, in combination with an apparent predator-driven increase in cryptic behavior (Fig. 5), our results indicate that the threat of predation conferred by the presence of large predators may indirectly constrain sea urchin grazing to only the most spatially complex reef habitat. In this habitat individuals can briefly leave crevices to graze but may quickly return to proximal refuge upon threat of predation (e.g., Tegner and Dayton 1981, Cowen 1983, Tegner and Levin 1983, Sala and Zabala 1996; reviewed by Sala et al. [1998]).

The potential ecological importance of predator-induced cryptic behavior among diademid urchins was first suggested by observations of Fricke 1974 (cited in Nelson and Vance [1979]), who described that grazing by the homing *Diadema setosum* (which demonstrates the usual nocturnal Diademid foraging pattern) occurred both night and day when day-active predatory fish were locally absent. Indeed, patterns of dispersion, local distribution, and nocturnal behavior of *C. rodgersii* in New South Wales have been inferred as the result of predator avoidance of the large day-active predator (on *C. rodgersii*) *Achoerodus viridis* (Labridae) (reviewed by Andrew and Byrne [2007]). A shifted ecological baseline is suggested by report of the large urchin-eating wrasse *Achoerodus viridis* (Dayton et al. 1998), which was anecdotally common in Tasmania in the 1920s, and was reported to be marketed as adults exceeding 1 m in length in the 1800s (Lord and Scott 1924, cited in Last et al. 2010), but was likely driven to local extinction by intensive inshore gillnet fishing. Not seen on the Tasmanian coast for 50 years (see Last et al. 2010), a pair of juvenile *A. viridis* (300 mm TFL; unlikely to be effective urchin predators at this size [Gillanders 1995]) were observed over several occasions to be residing locally on a *C. rodgersii* barren in northeast Tasmania in 2004; however, these individuals subsequently disappeared coincident with a period of intensive gillnetting (S. D. Ling, *personal observation*).

While the apparent functional redundancy of sea urchin predators within the range extension region of *Centrostephanus rodgersii* appears very low, the nocturnally active heterodontid shark *Heterodontus portusjacksoni* (McLaughlin and O'Gower 1971) is also known to consume *C. rodgersii*. However, this predominately warmer water New South Wales species is also relatively rare on rocky reefs in Tasmania, and is also highly susceptible to gillnet fishing. Ultimately, the presence of entire suites of naturally large predators, including both lobsters and fish, as restorable by long-term protection within marine reserves (e.g., Byrnes et al. 2006, Edgar et al. 2009), would appear to confer maximum resilience of kelp beds against *C. rodgersii* overgrazing by acting to restrict grazing activity to the most spatially complex reef habitats, thus minimizing the risk, extent, persistence, and types of habitats on which widespread barrens may manifest. Other than no-take marine reserve protection, rotational spatial closures, managed predator relocation, upper size limits and/or catch reductions, allowing accumulation of large predator biomass on rocky reefs, appear to be the only options for rebuilding the biomass of large and thus functional sea urchin predators, given their high vulnerability to fishing.

CONCLUSIONS

Our experimental results clearly show that size and habitat-specific factors define the survival of *Centrostephanus rodgersii* within the sea urchins' range extension



PLATE 1. Photograph of the range-extending sea urchin *Centrostephanus rodgersii* forming patchy barrens among a kelp bed in southeast Tasmania in 2009 at a depth of 6 m. The photograph shows an individual sea urchin exposed on a flat rock surface (boulder top in foreground) and an additional four individuals sheltering among boulders (background). The kelp bed habitat is composed of *Ecklonia radiata* (Order Laminariales) and *Phyllospora comosa* (Order Fucales). Photo credit: S. D. Ling.

region. This indicates that ecosystem impacts mediated by the climate-driven range extension of habitat-modifying organisms will be heterogeneous across different habitats, and that the likelihood and spatial extent of such ecosystem-level impacts will, all else being equal, be greater where such populations experience minimal predation risk within such novel systems. Given strong local recovery of large predators inside marine reserves within the range extension region of *C. rodgersii*, we have shown that marine reserves are indeed an effective local-scale tool capable of reinstating trophic dynamics across reef habitats, ultimately increasing risk of predation for sea urchins and thus decreasing the likelihood and extent of catastrophic overgrazing in the face of ongoing ocean warming. Finally, our findings highlight the crucial need to account for the influence of size and habitat dynamics on rates of key predator–prey interactions when assessing the performance of marine reserves, and when

ultimately managing stakeholder expectations of ecosystem-level responses within marine reserve boundaries.

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